

Phase Gate Review Checklist

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System

Type: Hotel Reservation

Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)

Date: February 20, 2026

Version: 1.0

Phase Gate 1: Requirements → Design

Objective: Ensure requirements are complete, clear, and approved before starting design work.

Entry Criteria

- ☒ Stakeholder interviews completed (Elena Petrova, Hotel Manager)
- ☒ Requirements gathering notes compiled from stakeholder meeting
- ☒ Team understands project scope (hotel reservation system for Grand Vista Hotel)

Exit Criteria (Must Pass All)

- ☒ Requirements Specification Document completed
- ☒ All 12 functional requirements documented (FR-1 through FR-12)
- ☒ 5 non-functional requirements documented (NFR-1 through NFR-5)
- ☒ Requirements are clear, testable, and traceable
- ☒ Scope is defined (in-scope and out-of-scope items listed)
- ☒ Stakeholder has reviewed requirements
- ☒ No major ambiguities or missing information

Review Checklist

Completeness

- ☒ Requirements cover all stakeholder needs: room search, booking, cancellation, modification, check-in/out, staff dashboard, reporting
- ☒ Both functional and non-functional requirements included
- ☒ Assumptions and constraints documented (budget, timeline, team size, technology)

Quality

- ☒ Requirements written clearly with measurable criteria (no vague terms)
- ☒ Requirements are testable (each FR has corresponding acceptance criteria)
- ☒ Requirements prioritized (High / Medium / Low)

Stakeholder Alignment

- ☒ Stakeholder (Elena Petrova) reviewed and approved the requirements
- ☒ No conflicting or unclear requirements requiring resolution

Sign-Off

Prepared By: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)

Date: February 20, 2026 (End of Requirements Activity)

Stakeholder/Reviewer Approval: _____ Date: _____

Decision: ☒ Approved - Proceed to Design | ☐ Revisions Needed

Comments/Feedback: All 12 functional requirements documented with clear priorities. Non-functional requirements include measurable performance targets. Scope is well-defined with realistic out-of-scope items for V1. Approved to proceed to design phase.

Phase Gate 2: Design → Implementation

Objective: Ensure design is sound, complete, and approved before starting implementation.

Entry Criteria

- ☒ Phase Gate 1 passed (requirements approved)
- ☒ Requirements Specification Document available and referenced
- ☒ Team understands system goals and constraints

Exit Criteria (Must Pass All)

- ☒ Design Documentation completed
- ☒ System architecture diagram created (three-tier: React, Express, PostgreSQL)
- ☒ Component specifications documented (Frontend, Backend API, Database)
- ☒ Data model defined (Users, Rooms, Reservations entities with attributes and relationships)
- ☒ Technology stack selected and justified
- ☒ Design addresses all 12 functional requirements
- ☒ Design reviewed and approved

Review Checklist

Completeness

- ☒ Design covers all 12 functional requirements with corresponding API endpoints and UI views
- ☒ All major components identified: Authentication, Room Management, Reservation, Notification, Reporting modules
- ☒ Data model complete: Users, Rooms, Reservations entities with all attributes and foreign keys

Quality

- ☒ Architecture follows modularity principle (separate frontend, backend, database layers)
- ☒ Component interfaces clearly defined (12 REST API endpoints specified)
- ☒ Technology choices appropriate and justified (PostgreSQL for ACID transactions, React for SPA)

Traceability

- ☒ Design components trace back to requirements (each FR maps to API endpoint and UI view)
- ☒ Each requirement has a corresponding design element

Feasibility

- ☒ Design is realistic to implement within 4-week implementation phase
- ☒ Technical risks identified: date overlap handling, concurrent booking prevention

Sign-Off

Prepared By: Danylo Chernov (Design Lead)

Date: February 20, 2026 (End of Design Activity)

Stakeholder/Reviewer Approval: _____ Date: _____

Decision: ☒ Approved - Proceed to Implementation | ☐ Revisions Needed

Comments/Feedback: Architecture is well-structured with clear separation of concerns. Data model correctly captures the core hotel reservation domain. API endpoints cover all required functionality. PostgreSQL choice is well-justified for booking integrity. Approved to proceed.

Phase Gate 3: Testing → Reflection

Objective: Ensure test plan is complete, comprehensive, and requirements are traced to test cases.

Entry Criteria

- ☒ Phase Gate 2 passed (design approved)
- ☒ Design Documentation available
- ☒ Requirements Specification Document available for traceability

Exit Criteria (Must Pass All)

- ☒ Test Plan completed with 15 test cases covering functional and non-functional requirements
- ☒ Requirements Traceability Matrix maps all requirements to test cases
- ☒ Test strategy defined (unit, integration, system, UAT levels)
- ☒ Entry and exit criteria for testing phase documented
- ☒ Defect management process and severity levels defined
- ☒ Test schedule and environment specified

Review Checklist

Test Coverage

- ☒ All 12 functional requirements have at least one test case
- ☒ Non-functional requirements (performance, usability, security) have dedicated test cases
- ☒ Requirements Traceability Matrix complete: 14/17 requirements have dedicated test cases

Quality

- ☒ Test cases include clear preconditions, steps, and expected results
- ☒ Test cases cover both positive and negative scenarios (valid login + invalid login)
- ☒ Edge cases addressed: double booking prevention, cancellation time limits, date overlaps

Readiness

- ☒ Test environment specifications documented
- ☒ Test data requirements defined (50 rooms, 200 reservations, 50 users)
- ☒ Team roles assigned for testing activities

Sign-Off

Prepared By: Daniil Ustiuhov (QA Lead)

Date: February 20, 2026 (End of Test Plan Activity)

Stakeholder/Reviewer Approval: _____ Date: _____

Decision: ☒ Approved - Proceed to Reflection | ☐ Revisions Needed

Comments/Feedback: Test plan is thorough with good coverage of core reservation workflows. Traceability matrix demonstrates systematic mapping of requirements to tests. Both positive and negative test cases included. The double-booking prevention test (TC-06) is particularly important for system integrity. Approved to proceed to process reflection.

Summary: Phase Gate Quick Reference

Gate	Transition	Key Deliverable	Key Question	Status
PG1	Requirements → Design	Requirements Spec	Are requirements complete and approved?	✓ Passed
PG2	Design → Implementation	Design Documentation	Is design sound and ready to build?	✓ Passed
PG3	Testing → Reflection	Test Plan	Is test plan complete with traceability?	✓ Passed